

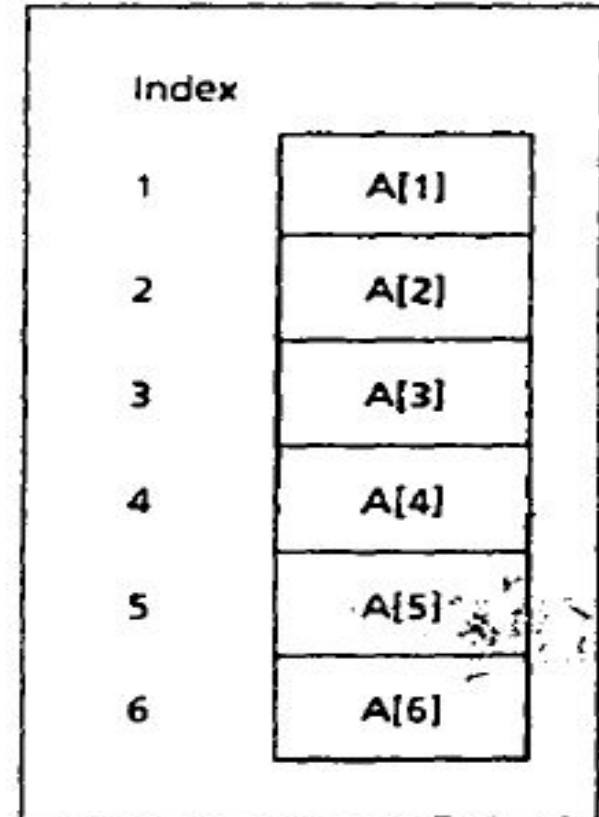
CSE 315

Microprocessors, Microcontrollers, and
Embedded Systems

Arrays and Addressing Modes

One Dimensional Array

- A one dimensional array is an ordered list of elements
- A 5 character string named MSG
MSG DB 'abcde'
- A word array W of 6 integers
W DW 10, 20, 30, 40, 50, 60



One Dimensional Array

- The address of the array variable is called the base address of the array
- The array W is like this

<i>Offset address</i>	<i>Symbolic address</i>	<i>Decimal content</i>
0200h	W	10
0202h	W+2h	20
0204h	W+4h	30
0206h	W+6h	40
0208h	W+8h	50
020Ah	W+Ah	60

DUP operator

repeat_count DUP (value)

- An array of 100 words, each initialized to 0

GAMMA DW 100 DUP (0)

- An array of 212 uninitialized bytes.

DELTA DB 212 DUP (?)

- Nested DUP operator

LINE DB 5, 4, 3 DUP (2, 3 DUP (0), 1)

is equivalent to

LINE DB 5, 4, 2, 0, 0, 0, 1, 2, 0, 0, 0, 1, 2, 0, 0, 0, 1

Location of Array Elements

- A is an array of N elements
- S is the number of bytes in an element of the array
 - $S = 1$ (for byte array)
 - $= 2$ (for word array)
- So, the location of the elements of A can be determined as

Position	Location
1	A
2	$A = 1 \times S$
3	$A = 2 \times S$
.	...
.	.
.	.
N	$A = (N - 1) \times S$

Addressing Modes

- So far we have used
 1. Register mode operand is a register
 2. Immediate mode operand is constant
 3. Direct mode operand is a variable
- MOV AX, 0 ; Destination AX is register mode, source 0 is immediate mode
- ADD X, AX ; Destination X is direct mode, source AX is register mode
- There are four additional addressing modes
 1. Register indirect mode
 2. Based mode
 3. Indexed mode
 4. Based indexed mode

Register Indirect Mode

- The offset address of the operand is contained in a register.
- Operand format [register]
- Register can be BX, SI, DI, or BP.

- If BX, SI, or DI contains the offset of the operand,
DS contains the segment number
- If BP contains the offset of the operand,
SS contains the segment number

Register Indirect Mode

Write a code to sum in AX the elements of the 10-element array W defined by
W DW 10, 20, 30, 40, 50, 60, 70, 80, 90, 100

```
XOR AX, AX    ; AX holds sum
LEA SI, W      ; SI points to array W

MOV     CX, 10 ; CX has number of elements
ADDNOS:
  ADD AX, [SI] ; sum=sum+element
  ADD     SI, 2 ; move pointer to the next element

  LOOP   ADDNOS ; loop until done
```

Based Mode

- Operand format [register + displacement]
[displacement + register]
[register] + displacement
displacement + [register]
displacement [register]
- Register can be BX (base register), or BP (base pointer)
- Displacement can be
 - the offset address of a variable (e.g., A)
 - a constant (positive or negative) (e.g., -2)
 - the offset address of a variable plus or minus a constant (A + 2)
- If BX is used as register, DS contains the segment number
- If BP is used as register, SS contains the segment number

Based Mode

- Suppose W is a word array and BX contains 4
- Displacement is the offset address of variable W.
- So, to move the 3rd element of the array to AX, we can write

MOV AX, [BX + W]	;[register + displacement]
MOV AX, [W + BX]	;[displacement + register]
MOV AX, [BX] + W	;[register] + displacement
MOV AX, W + [BX]	;displacement + [register]
MOV AX, W [BX]	;displacement [register]

Indexed Mode

- Operand format [register + displacement]
[displacement + register]
[register] + displacement
displacement + [register]
displacement [register]
- Register can be SI (source index), or DI (destination index)
- Displacement can be
 - the offset address of a variable (e.g., A)
 - a constant (positive or negative) (e.g., -2)
 - the offset address of a variable plus or minus a constant (A + 2)
- If SI, or DI is used as register, DS contains the segment number

PTR operator

- Instructions such as

`MOV [BX], 1`

is not allowed since whether the destination is byte or word is unknown

- The PTR operator can be used for type casting

`MOV     BYTE PTR [BX], 1`

`MOV     WORD PTR [BX], 1`